

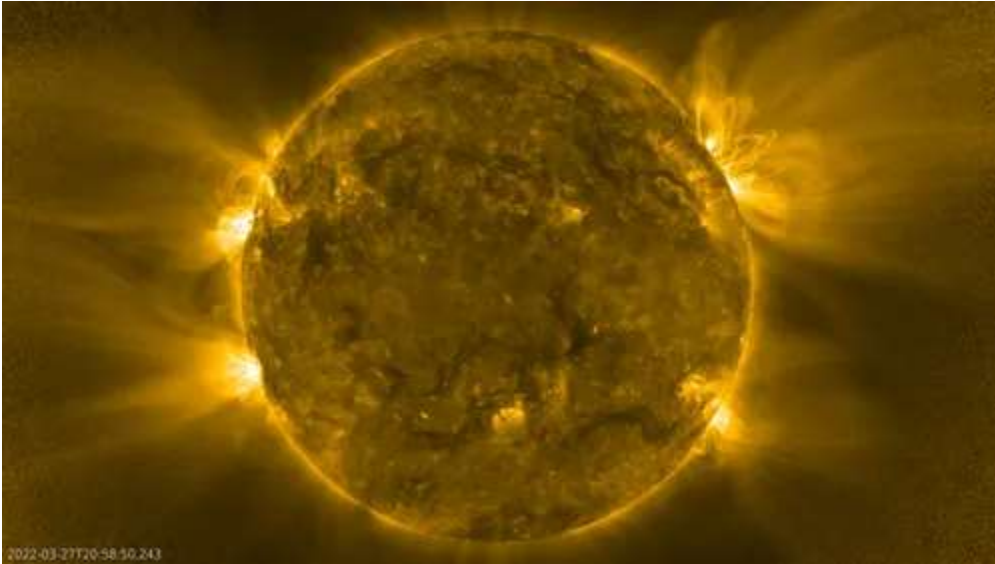
Solar Flares

Defination:-

Solar flares are sudden, intense bursts of electromagnetic radiation and high-energy particles erupting from the Sun's atmosphere, typically occurring in active regions near sunspots where magnetic fields are strongest.

Solar flares originate from a process called magnetic reconnection, where twisted and sheared magnetic field lines in the Sun's corona suddenly break and reconnect into a lower-energy configuration.





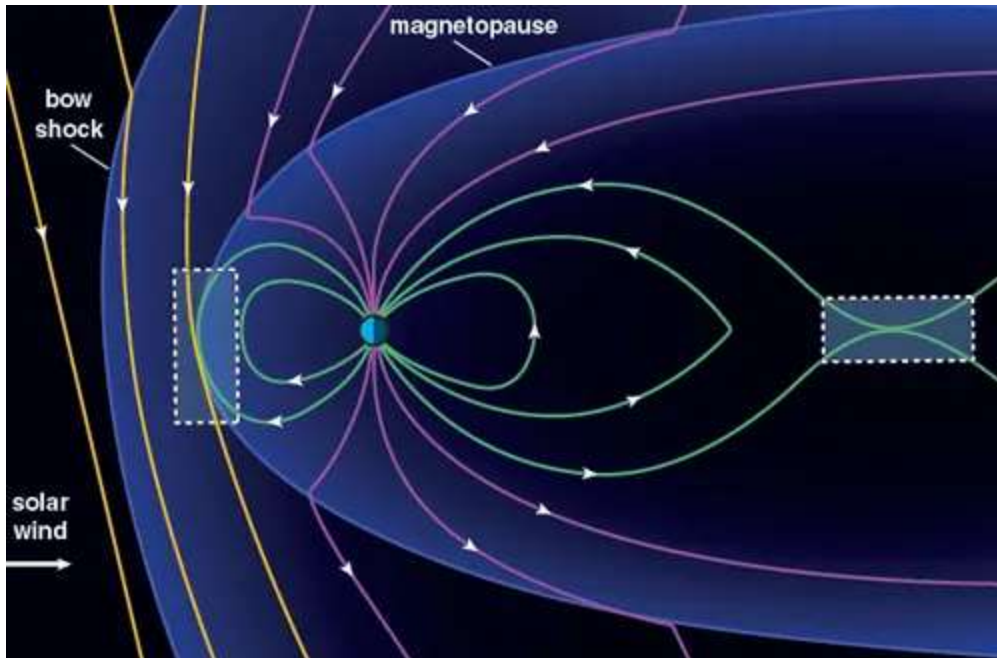
As seen in the above images it ejects intense bursts of electromagnetic radiation including X-rays and Gamma rays.

Formation of solar flares:-

Magnetic Reconnection Mechanism

The fundamental driver of solar flares is magnetic reconnection. Diagrams typically show twisted magnetic field lines approaching each other, breaking at an "X-point," and reconnecting into a lower-energy configuration.

This process creates a vertical current sheet beneath an erupting flux rope, launching high-speed plasma jets away from the reconnection site and heating the surrounding plasma to millions of degrees.

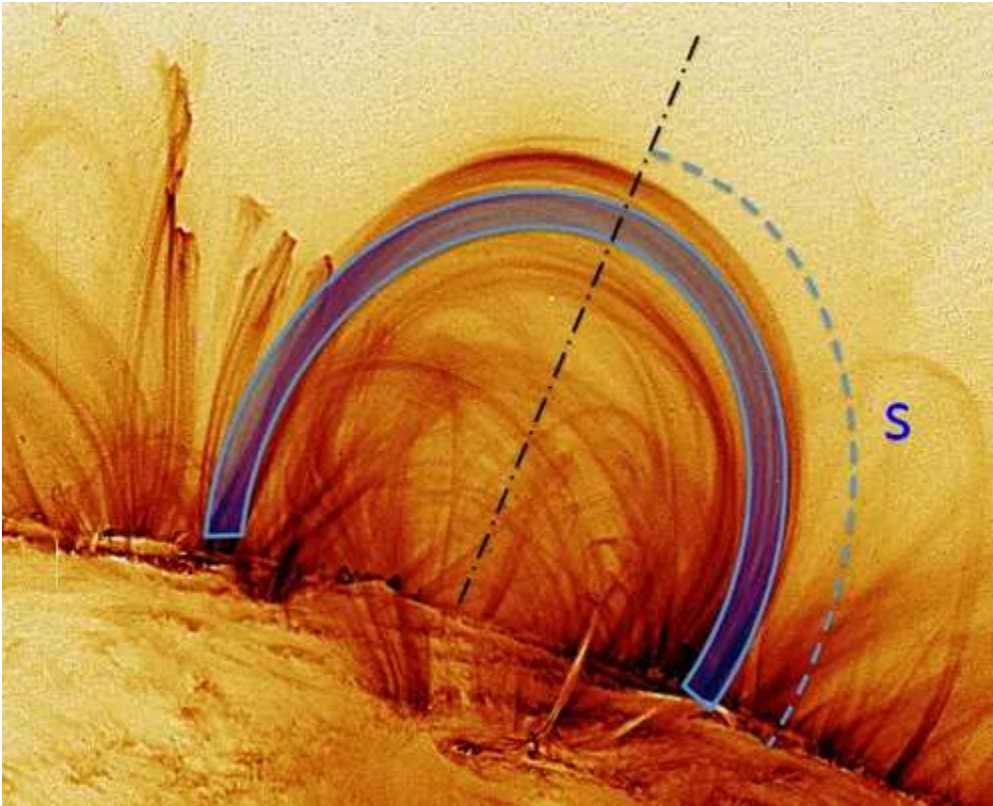


Anatomy of a Solar Flare

Flares generally have the following components:

- **Flare Ribbons:** Bright loops anchored in the photosphere where accelerated particles impact the dense lower atmosphere.
- **Post-Flare Loops:** Arcades of hot plasma formed as reconnected field lines snap back.
- **Cusp Structure:** A distinctive wedge-shaped region of hot plasma located directly above the loops, marking the site of ongoing reconnection.

- Erupting Filament: A dark coil of cooler plasma lifted into the corona, often preceding the flare.



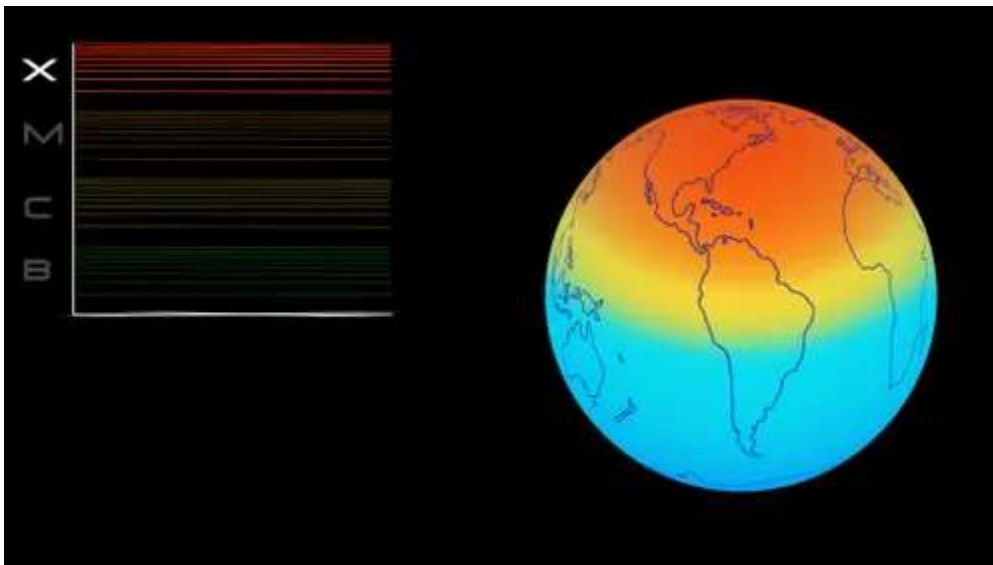
Classification of Solar Flares:-

Flares are categorized by their peak X-ray flux (measured in watts per square meter) into five logarithmic classes, where each step represents a tenfold increase in energy.

In each letter class there is a finer scale from 1-9.

Although in X class there are flares more than 10 times the power of x1. So X class flares can go higher than 9.

- *A & B Classes*: Background levels; too weak to affect Earth.
- *C Class*: Minor; rarely cause noticeable effects.
- *M Class*: Moderate; can cause brief radio blackouts at polar regions and minor radiation storms threatening astronauts.
- *X Class*: Extreme; the most powerful eruptions. These are numbered 1–9 (e.g., X2 is twice as intense as X1), but the scale is open-ended. The strongest recorded flare was an X28 in 2003, which saturated sensors. X-class flares trigger global radio blackouts (R3–R5 scale) and severe geomagnetic storms.



Class	W/m ² between 1 & 8 Ångströms
A	$<10^{-7}$
B	$\geq 10^{-7} < 10^{-6}$

C	$\geq 10^{-6} < 10^{-5}$
M	$\geq 10^{-5} < 10^{-4}$
X	$\geq 10^{-4}$

Major Solar Flares in Recorded History

The most significant solar flares are ranked by their estimated X-ray intensity.

While modern instruments have directly measured flares since 1976, historical events are estimated based on geomagnetic data and isotopic records.

1. The Carrington Event (September 1, 1859)

Estimated Magnitude: ~X45 to X64 (Revised estimates).

Significance: The most powerful geomagnetic storm in recorded history. Observed visually by Richard Carrington and Richard Hodgson as a white-light flare.

Impact: The associated CME reached Earth in just 17.6 hours, causing telegraph systems to spark, catch fire, and operate without power. Auroras were visible as far south as the Caribbean and Colombia. If it occurred today, it would cause catastrophic grid failure globally.

2. The Halloween Storms (October–November 2003)

Key Event: X28 Flare (November 4, 2003).

Estimated Magnitude: Later analysis suggests the peak may

have reached X45, rivaling the Carrington flare's intensity, though it was partially limb-occulted.

Significance: The most intense flare directly measured by modern satellites (GOES sensors saturated at X17).

Impact: Caused widespread satellite anomalies (including the loss of Japan's ADEOS-II), forced astronauts on the ISS to shelter, and created auroras visible in Florida and Texas.

3. The Quebec-Causing Flare (March 1989)

Magnitude: X15 (March 10, 1989).

Significance: Triggered the most famous modern geomagnetic disaster.

Impact: The resulting geomagnetic storm collapsed the Hydro-Québec power grid in Canada within 90 seconds, leaving 6 million people without electricity for 9 hours. It also caused the loss of control of the TDRS-1 communications satellite.

4. The Bastille Day Event (July 14, 2000)

Magnitude: X5.7 (Some estimates up to X6).

Significance: A massive eruption occurring on France's national holiday.

Impact: Produced a severe radiation storm (S3) and a fast CME that caused a G5 geomagnetic storm. It is notable for the "three-ribbon" flare structure observed and significant disruption to GPS and radio communications.

5. Largest Flares of Solar Cycle 25 (2024–2026)

As of June 2026, Solar Cycle 25 has produced several historic events, surpassing initial predictions:

X8.1 / X8.3 Flare (February 1–2, 2026): One of the strongest of the current cycle, emitted by active region AR4366. It caused R3-level radio blackouts across the Pacific and Australia.

X5.16 Flare (November 11, 2025): The strongest flare of 2025, emitted by region AR4274.

X2.5 Flare (April 2026): A significant late-cycle event contributing to the cycle's "double peak" behavior.